

Johnny To

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SUMMARY

Recent MS graduate in Software Engineering with a specialization in Data Science from SJSU, with project experience spanning LLM systems, RAG pipelines, and full-stack mobile and web development. Looking to bring both software engineering depth and applied AI skills to a collaborative engineering team.

EDUCATION

San Jose State University

MS in Software Engineering, Specialization in Data Science

San Jose, California

August 2024 – May 2026

- **Relevant Coursework:** Enterprise Software Platforms, Software Systems Engineering, Software QA Testing

University of California, Riverside

BS in Computer Science

Riverside, California

September 2021 – December 2024

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Frameworks: React, React Native, Next.js, Express.js

AI/ML: LangChain, RAG, Prompt Engineering, LLM Benchmarking, FAISS, Ollama, OpenAI API

Data Science: NumPy, Pandas, Jupyter Notebooks

Cloud/DevOps: AWS (Lambda, API Gateway), Docker

Databases: PostgreSQL, MongoDB

Tools: Git, GitHub

PROJECTS

SJSU Course & Degree Virtual Assistant | *Python, LangChain, FAISS, OpenAI API* | [GitHub](#)

- Architected an LLM-based agent system integrating tool orchestration, multi-source retrieval (RAG), and query planning to answer SJSU students' course planning and degree audit questions.
- Built a dynamic tool routing pipeline enabling the model to select between database queries, retrieval systems, and constrained web search.
- Benchmarked Qwen 1.7B vs 14B models, identifying latency/quality tradeoffs (20–30s vs 60–100s inference).
- Designed secure tool-scoped data access, preventing leakage of sensitive student information under adversarial prompts.

LessGo: Smart Carpooling Platform | *Python, Node.js, React Native, PostgreSQL, FastAPI* | [GitHub](#)

- Architected a full-stack microservice carpooling platform for SJSU students, spanning 11 independent services including matching, routing, payments, and real-time safety monitoring.
- Implemented a three-stage rider-driver matching pipeline combining PostGIS proximity filtering, RShareForm heterogeneous network embeddings, and a weighted service cost function, achieving 84%+ match accuracy across candidate pools of 100 drivers.
- Engineered a real-time safety monitoring system that detects route deviations and speed anomalies by comparing live GPS updates against planned route polylines, triggering automated alerts on suspicious driver behavior.

EXPERIENCE

UCR Bourns College of Engineering

Undergraduate Research Assistant

Riverside, California

June 2021 – August 2021

- Developed a NumPy-based benchmarking framework to evaluate the performance of experimental matrix multiplication methods.
- Reduced computational steps by 33% through algorithmic optimization.
- Collaborated with faculty to test novel approaches to accelerating deep learning workloads.

LEADERSHIP

Developers' Guild

Vice President

Cupertino, California

2019 – 2021

- Led a redesign of club assets resulting in an 88% increase in new student sign-ups.
- Taught Python and Web Development to club members; coordinated the school's first Hacktoberfest event.